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(57) Abstract :

ABSTRACT WEBCAM-BASED EYE GAZE AND VOICE CONTROLLED MOUSE INTERFACE SYSTEM FOR MOTOR-IMPAIRED USERS The present invention provides a webcam-based eye gaze and voice-controlled mouse interface system (100) for hands-free computer operation by motor-impaired users. The system (100) comprises an eye tracking module (110) that captures webcam frames, detects iris centre coordinates through facial landmark detection (114), maps iris position through zone-based coordinate transformation (116), applies exponential smoothing (118), and injects smoothed coordinates into the operating system cursor control interface (120). A voice control module (130) operates concurrently on a separate thread, capturing microphone audio, performing speech recognition (134), matching commands through substring-based dispatch (136), and executing actions through operating system event injection (138). A Gmail control module (140) enables browser-agnostic email management through keyboard shortcut injection. The system is characterized in that cursor movement and action execution operate on separate independent input channels, eliminating false positive clicks from involuntary blinking. [to be published with fig.1]

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